SPRAWOZDANIE

Zajęcia: Grafika i gry komputerowe I
Prowadzący: mgr. inż. Jakub Krawczyk

Laboratorium 7
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Temat:
„Grafika 3D w bibliotece WebGL/GLSL
”

Sprawozdanie :

https://github.com/Kamalo00/Grafika-komputerowa-pliki.git

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ZADANIE 1

1. Polecenie :

Plik lab12.html pokazuje mały sześcian, który można obrócić, przeciągając myszą na płótnie. Zadaniem jest zastąpienie sześcianu dużym wiatrakiem siedzącym na prostokątnej podstawie, jak pokazano na rysunku. Łopatki wiatraka powinny obracać się po włączeniu animacji. Każda łopatka wiatraka powinna być zbudowana z dwóch stożków. (Dodanie czajniczka, który znajduje się na podstawie, jest konieczne dla uzyskania oceny "5")

Program zawiera trzy zmienne instancji reprezentujące podstawowe obiekty: cube, cone, cylinder. Te zmienne mają metody instancji cube.render(), cone.render(), cylinder.render(), które można wywołać w celu narysowania obiektów. Obiekty nietransformowane mają rozmiar 1 we wszystkich trzech kierunkach i mają swój środek na (0,0,0). Oś stożka i oś cylindra są wyrównane wzdłuż osi Z. Wszystkie obiekty na scenie powinny być przekształconymi wersjami podstawowych obiektów (lub podstawowego obiektu czajnika).

2. Wprowadzane dane:

Celem zadania było stworzenie turbiny wiatrowej z wykorzystaniem stożków , elementów animacji oraz dodanie czajnika w podstawie wiatraka.

3. Wykorzystane komendy:

```
"use strict";

var gl;   // The webgl context
var canvas;    // The canvas where gl draws

var a_coords_loc;         // Location of the a_coords attribute variable in the shader program
var a_normal_loc;         // Location of a_normal attribute

var u_modelview;       // Locations for uniform matrices
var u_projection;
var u_normalMatrix;

var u_material;     // An object tolds uniform locations for the material.
var u_lights;       // An array of objects that holds uniform locations for light properties.

var projection = mat4.create();    // projection matrix
var modelview = mat4.create();    // modelview matrix
var normalMatrix = mat3.create();  // matrix for transforming normal vectors

var frameNumber = 0;  // frame number during animation

var cone, cylinder, cube;  // Basic objects, created using function createModel and basic-object-models-IFS.js.
                           // The cube is 1 unit on each side and is centered at (0,0,0).
                           // the cone and cylinder have diameter 1 and height 1 and are centered at
                           //     (0,0,0), with their axes aligned along the z-axis.

var matrixStack = [];           // A stack of matrices for implementing hierarchical graphics

var currentColor = [1,1,1];   // The current diffuse color; render() functions in the basic objects set
                              // the diffuse color to currentColor when it is called before drawing the object
                              // Specular color properties, which don't change, are set in initGL()

var rotateX = 0, rotateY = 0;   // Overal rotation of model, in radians, set by mouse dragging.

/**
 * Draws the image, which consists of either the "world" or a closeup of the "car".
 */
```

```
function addBlade()
{
    pushMatrix();
    currentColor = [200/255, 1, 252/255];
    mat4.rotateY(modelview, modelview, Math.PI /2);
    mat4.translate(modelview, modelview, [0, 0, 2.7]);
    mat4.scale(modelview, modelview, [0.7, 0.7, 4.1]);
    cone.render();
    popMatrix();

    pushMatrix();
    currentColor = [200/255, 1, 252/255];
    mat4.translate(modelview, modelview, [0.3, 0, 0]);
    mat4.rotateY(modelview, modelview, Math.PI /2);
    mat4.rotateX(modelview, modelview, Math.PI );
    mat4.scale(modelview, modelview,[0.7, 0.7, 0.7]);
    cone.render();
    popMatrix();

    }

var rotateEachFrame = 0.5;
let wierzcholki = 3;
```

```
function draw() {
    gl.clearColor(0,0,0,1);
    gl.clear(gl.COLOR_BUFFER_BIT | gl.DEPTH_BUFFER_BIT);


    mat4.perspective(projection, Math.PI/4, 1, 1, 50);
    gl.uniformMatrix4fv(u_projection, false, projection );

    mat4.lookAt(modelview, [0,0,25], [0,0,0], [0,1,0]);

    mat4.rotateX(modelview,modelview,rotateX);
    mat4.rotateY(modelview,modelview,rotateY);

    pushMatrix();
    currentColor = [0.999, 0.111, 0.111];
    mat4.translate(modelview, modelview, [0, -5, 0]);
    mat4.scale(modelview,modelview,[5, 1, 5]);
    cube.render();
    popMatrix();


    pushMatrix();
    currentColor = [0.999, 0.555, 0.333];
    mat4.translate(modelview, modelview, [0, 0, 0]);
    mat4.rotateX(modelview,modelview, Math.PI * 0.5);
    mat4.scale(modelview,modelview,[0.4, 0.4, 10]);
    cylinder.render();
    popMatrix();


    pushMatrix();
    currentColor = [0.222, 0.888, 0.222];
    mat4.translate(modelview, modelview, [1.2, -4.1, 2.0]);
    mat4.scale(modelview,modelview,[0.05, 0.05, 0.05]);
    teapot.render();
    popMatrix();

    pushMatrix();
    mat4.translate(modelview, modelview, [-2.9, 2, 0.2]);
    pushMatrix();
    mat4.translate(modelview, modelview, [2.9, 2, 0]);
    mat4.rotate(modelview, modelview, rotateEachFrame, [0, 0, 1]);
    mat4.translate(modelview, modelview, [2.9, -2, 0]);

    for(let i =0; i<wierzcholki;i++){

        pushMatrix();
        mat4.translate(modelview, modelview, [-3, 1.95, 0]);
        mat4.rotateZ(modelview,modelview, i*(360/wierzcholki)*(Math.PI/180));
        mat4.rotateY(modelview,modelview, Math.PI);
        addBlade();
        popMatrix();

    }
} // end draw();
```

```
function pushMatrix() {
    matrixStack.push( mat4.clone(modelview) );
}


/**
 *  Restore the modelview matrix to a value popped from the matrix stack.
 */
function popMatrix() {
    modelview = matrixStack.pop();
}
```

```
function createModel(modelData) {
    var model = {};
    model.coordsBuffer = gl.createBuffer();
    model.normalBuffer = gl.createBuffer();
    model.indexBuffer = gl.createBuffer();
    model.count = modelData.indices.length;
    gl.bindBuffer(gl.ARRAY_BUFFER, model.coordsBuffer);
    gl.bufferData(gl.ARRAY_BUFFER, modelData.vertexPositions, gl.STATIC_DRAW);
    gl.bindBuffer(gl.ARRAY_BUFFER, model.normalBuffer);
    gl.bufferData(gl.ARRAY_BUFFER, modelData.vertexNormals, gl.STATIC_DRAW);
    gl.bindBuffer(gl.ELEMENT_ARRAY_BUFFER, model.indexBuffer);
    gl.bufferData(gl.ELEMENT_ARRAY_BUFFER, modelData.indices, gl.STATIC_DRAW);
    model.render = function() {  // This function will render the object.
            // Since the buffer from which we are taking the coordinates and normals
            // changes each time an object is drawn, we have to use gl.vertexAttribPointer
            // to specify the location of the data. And to do that, we must first
            // bind the buffer that contains the data.  Similarly, we have to
            // bind this object's index buffer before calling gl.drawElements.
        gl.bindBuffer(gl.ARRAY_BUFFER, this.coordsBuffer);
        gl.vertexAttribPointer(a_coords_loc, 3, gl.FLOAT, false, 0, 0);
        gl.bindBuffer(gl.ARRAY_BUFFER, this.normalBuffer);
        gl.vertexAttribPointer(a_normal_loc, 3, gl.FLOAT, false, 0, 0);
        gl.uniform3fv(u_material.diffuseColor, currentColor);
        gl.uniformMatrix4fv(u_modelview, false, modelview );
        mat3.normalFromMat4(normalMatrix, modelview);
        gl.uniformMatrix3fv(u_normalMatrix, false, normalMatrix);
        gl.bindBuffer(gl.ELEMENT_ARRAY_BUFFER, this.indexBuffer);
        gl.drawElements(gl.TRIANGLES, this.count, gl.UNSIGNED_SHORT, 0);
        if (this.xtraTranslate) {
            popMatrix();
        }
    }
    return model;
}
```

```
function createProgram(gl, vertexShaderID, fragmentShaderID, attribute0) {
    function getTextContent( elementID ) {
            // This nested function retrieves the text content of an
            // element on the web page.  It is used here to get the shader
            // source code from the script elements that contain it.
        var element = document.getElementById(elementID);
        var node = element.firstChild;
        var str = "";
        while (node) {
            if (node.nodeType == 3) // this is a text node
                str += node.textContent;
            node = node.nextSibling;
        }
        return str;
    }
    try {
        var vertexShaderSource = getTextContent( vertexShaderID );
        var fragmentShaderSource = getTextContent( fragmentShaderID );
    }
    catch (e) {
        throw "Error: Could not get shader source code from script elements.";
    }
    var vsh = gl.createShader( gl.VERTEX_SHADER );
    gl.shaderSource(vsh,vertexShaderSource);
    gl.compileShader(vsh);
    if ( ! gl.getShaderParameter(vsh, gl.COMPILE_STATUS) ) {
        throw "Error in vertex shader:  " + gl.getShaderInfoLog(vsh);
     }
    var fsh = gl.createShader( gl.FRAGMENT_SHADER );
    gl.shaderSource(fsh, fragmentShaderSource);
    gl.compileShader(fsh);
    if ( ! gl.getShaderParameter(fsh, gl.COMPILE_STATUS) ) {
       throw "Error in fragment shader:  " + gl.getShaderInfoLog(fsh);
    }
    var prog = gl.createProgram();
    gl.attachShader(prog,vsh);
    gl.attachShader(prog, fsh);
    if (attribute0) {
       gl.bindAttribLocation(prog,0,attribute0);
    }
    gl.linkProgram(prog);
    if ( ! gl.getProgramParameter( prog, gl.LINK_STATUS) ) {
       throw "Link error in program:  " + gl.getProgramInfoLog(prog);
    }
    return prog;
}
```

```
/* Initialize the WebGL context.  Called from init() */
function initGL() {
    var prog = createProgram(gl,"vshader-source","fshader-source", "a_coords");
    gl.useProgram(prog);
    gl.enable(gl.DEPTH_TEST);

    /* Get attribute and uniform locations */

    a_coords_loc =  gl.getAttribLocation(prog, "a_coords");
    a_normal_loc =  gl.getAttribLocation(prog, "a_normal");
    gl.enableVertexAttribArray(a_coords_loc);
    gl.enableVertexAttribArray(a_normal_loc);

    u_modelview = gl.getUniformLocation(prog, "modelview");
    u_projection = gl.getUniformLocation(prog, "projection");
    u_normalMatrix =  gl.getUniformLocation(prog, "normalMatrix");
    u_material = {
        diffuseColor: gl.getUniformLocation(prog, "material.diffuseColor"),
        specularColor: gl.getUniformLocation(prog, "material.specularColor"),
        specularExponent: gl.getUniformLocation(prog, "material.specularExponent")
    };
    u_lights = new Array(4);
    for (var i = 0; i < 4; i++) {
        u_lights[i] = {
            enabled: gl.getUniformLocation(prog, "lights[" + i + "].enabled"),
            position: gl.getUniformLocation(prog, "lights[" + i + "].position"),
            color: gl.getUniformLocation(prog, "lights[" + i + "].color")
        };
    }

    gl.uniform3f( u_material.diffuseColor, 1, 1, 1 );  // set to white as a default.
    gl.uniform3f( u_material.specularColor, 0.1, 0.1, 0.1 );  // specular properties won't change
    gl.uniform1f( u_material.specularExponent, 32 );

    for (var i = 1; i < 4; i++) { // set defaults for lights
        gl.uniform1i( u_lights[i].enabled, 0 );
        gl.uniform4f( u_lights[i].position, 0, 0, 1, 0 );
        gl.uniform3f( u_lights[i].color, 1,1,1 );
    }

    // Set up lights here; they won't be changed.  Lights are fixed in eye coordinates.

    gl.uniform1i( u_lights[0].enabled, 1 );   // light is a "viewpoint light"
    gl.uniform4f( u_lights[0].position, 0,0,0,1 ); // positional, at viewpoint
    gl.uniform3f( u_lights[0].color, 0.6, 0.6, 0.6 );

    gl.uniform1i( u_lights[1].enabled, 1 );  // light 1 is a dimmer light shining from above
    gl.uniform4f( u_lights[0].position, 0,1,0,0 ); // diretionsl, from directino of positive y-axis
    gl.uniform3f( u_lights[0].color, 0.4, 0.4, 0.4 );

    gl.uniform1i( u_lights[2].enabled, 0 );  // lightes 2 and 3 are not used.

    // Note: position and spot direction for lights 1 to 4 are managed by modeling transforms.

} // end initGL()
```

```
function mouseDown(evt) {
    prevX = evt.clientX;
    prevY = evt.clientY;
    canvas.addEventListener("mousemove",mouseMove);
    document.addEventListener("mouseup",mouseUp);
    function mouseMove(evt) {
        var dx = evt.clientX - prevX;
        var dy = evt.clientY - prevY;
        rotateX += dy/200;
        rotateY += dx/200;
        prevX = evt.clientX;
        prevY = evt.clientY;
        draw();
    }
    function mouseUp(evt) {
        canvas.removeEventListener("mousemove",mouseMove);
        document.removeEventListener("mouseup",mouseUp);
    }
}


//-------------------------------- animation ----------------------------

var animating = false;

function frame() {
    if (animating) {
        rotateEachFrame=rotateEachFrame+Math.PI*0.01;
        frameNumber += 1;
        draw();

        requestAnimationFrame(frame);
    }
}

function setIsAnimating() {
    var run = document.getElementById("animCheck").checked;
    if (run != animating) {
        animating = run;
        if (animating)
            requestAnimationFrame(frame);
    }
}

//----------------------------------------------------------------------------


```

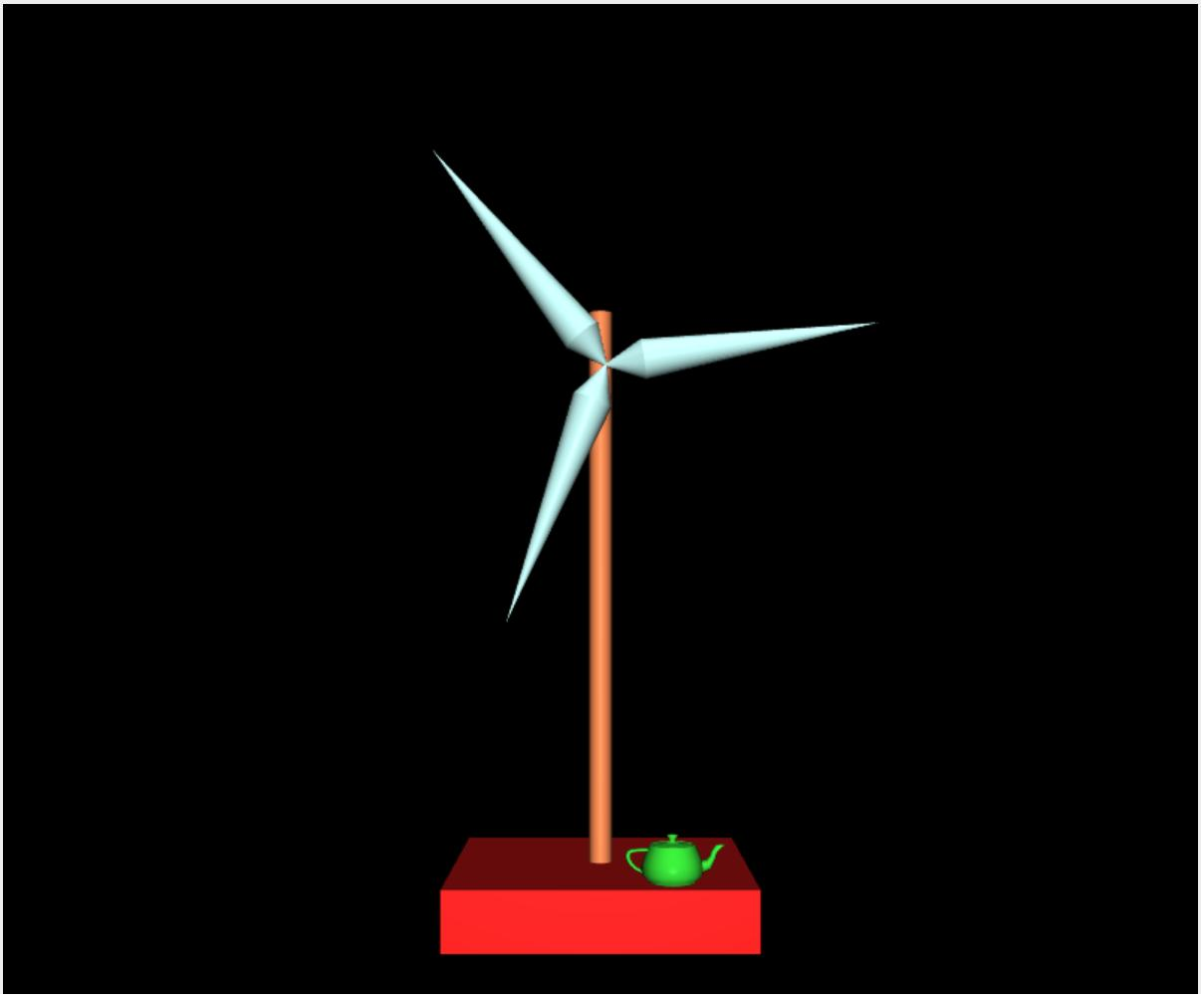
```
function init() {
    try {
        canvas = document.getElementById("webglcanvas");
        gl = canvas.getContext("webgl");
        if ( ! gl ) {
            throw "Browser does not support WebGL";
        }
    }
    catch (e) {
        document.getElementById("message").innerHTML =
            "<p>Sorry, could not get a WebGL graphics context.</p>";
        return;
    }
    try {
        initGL();  // initialize the WebGL graphics context
    }
    catch (e) {
        document.getElementById("message").innerHTML =
            "<p>Sorry, could not initialize the WebGL graphics context:" + e + "</p>";
        return;
    }
    document.getElementById("animCheck").checked = false;
    document.getElementById("animCheck").addEventListener("change", setIsAnimating);
    document.getElementById("reset").addEventListener("click", function() { rotateX = rotateY = 0; draw(); });
    canvas.addEventListener("mousedown", mouseDown);

    cone = createModel(uvCone());           // create the basic objects
    cylinder = createModel(uvCylinder());   //    uvCone(), uvCylinder(), and cube()
    cube = createModel(cube());             //    are defined in basic-object-models-IFS.js
    teapot = createModel(teapot());
    draw();
}
```

4. Wynik działania:

Zadanie WebGL3d

Przeciągnij myszkę w celu obracania modelu.

☑ Animuj Resetuj



5. Wnioski:

Hierarchia obiektów jest kluczowa dla budowy turbiny wiatrowej , ponieważ pozwala ona na grupowanie jej obiektów (m.in. stożka) co pozwala później na ustawienie wspólnej transformacji oraz pozycji dla tych elementów względem grupy. Ruch łopatek wiatraka uzyskuje się po przez zmianę właściwości jego obiektów w pętli renderowania.
Biblioteka Three.js ułatwia tworzenie scen 3D poprzez gotowe kształty (sześcian/walce/stożki itd.) oraz inne modele.